Ada’s Closet

1.1.0
9-13-26
Aaask
Ash Laufer
Alexis Hodges
Ava Jefferson
Shelby Flowers
Kelley Haina

Sandeep Sthapit

1.1 Purpose and Scope

The purpose of this document is to outline the details and processes of our project. The system that we will be designing is based on the 1995 film Clueless. The project is taking Sher's closet computer, that helps her pick out her outfit for the day. This system will help the end users picking out outfits that match.

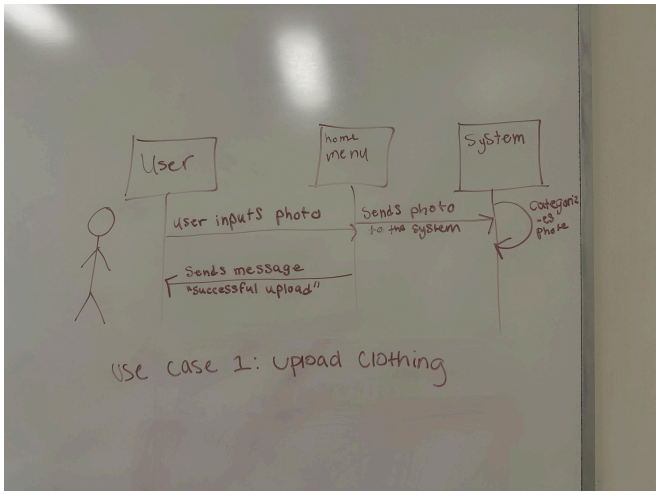
1.2 System Overview

Brief narrative description of the system being built.
Our system will be a virtual closet where users can upload images of their own clothes to receive outfit recommendations/ matches. Our system should be able to match clothes by color, style and weather.

Use-case model (diagram)

List of use cases (UC1, UC2, … UCn)
Each use case is a prerequisite of each use case beforehand. Each property is assumed for the next use case in order to function.

UC1: Upload Clothing



Preconditions (What needs to happen (in another use case) before this use case can start? What state must the system be in before the use case?):

- The user is in the application.
- The user has taken a photo of the clothing they want to upload.

Main Flow (The main flow is a series of declarative steps):

1. The user clicked the upload image button.
2. The user uploads the image into the application.
3. The application sends the image to the system
4. The system categorizes and adds a tag to the clothing item (i.e shortsleeve, long sleeve, pants, etc)
5. The user sees that the image was successfully uploaded into the application.

Subflows(Sub-flows break down the main flow and other sub-flows to improve document readability.):

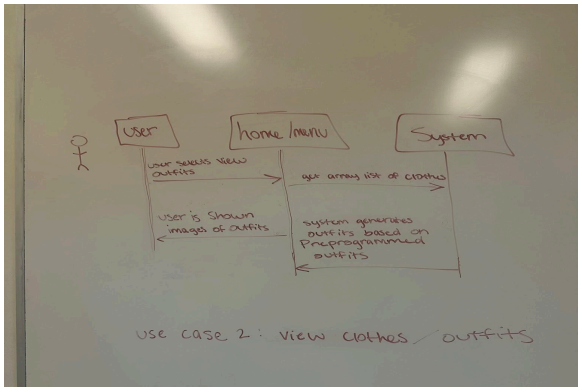
1. When the user is finding the image, the application needs to access the users camera roll and photos application.

Alternative Flows (The alternative flows define exceptional behavior that can interrupt the normal flow. Often alternative flows indicate what is to be done under error conditions.

To determine alternative flows, ask yourself, “What could possibly go wrong?” for each of the actions in the main flow and the sub-flow):

1. The user may accidentally upload the image multiple times which would cause nothing to happen.
2. The image the user uploads may not be compatible with the images accepted by the software. For example, the image does not originate from Samsung or Iphone default which are .img or .hic to be implemented in the software.

UC2: View Clothes



Preconditions:

- The user is in the application.
- The user has uploaded photos previously
- The user is looking through all the clothes in the user’s closet.
- The user has completed user case 1 at least once.

Main Flow (The main flow is a series of declarative steps):

1. The user clicks the “view” button that sends them to the screen that shows all the clothing in the closet of the user.
2. The application gets the array list of clothes uploaded
3. System generates outfits based on preprogrammed outfits
4. User is shown images of outfits

Subflows(Sub-flows break down the main flow and other sub-flows to improve document readability.):

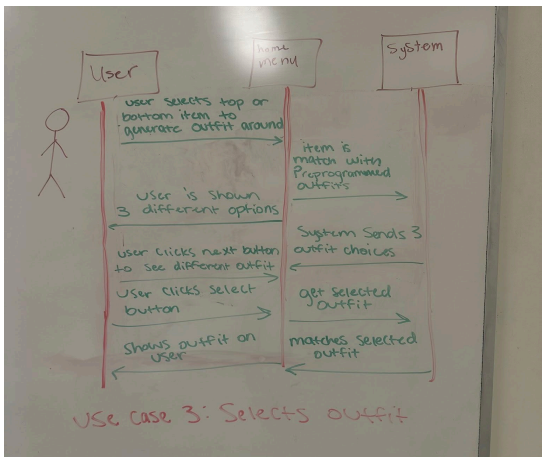
1. None

Alternative Flows (The alternative flows define exceptional behavior that can interrupt the normal flow. Often alternative flows indicate what is to be done under error conditions. To determine alternative flows, ask yourself, “What could possibly go wrong?” for each of the actions in the main flow and the sub-flow):

1. The user may not be able to find the button to access all the clothes in their closet.

UC3: Select Clothes

- The user picks a clothing item, the system will suggest three outfits that include the clothing item, and the user picks an outfit to view



Preconditions (What needs to happen (in another use case) before this use case can start? What state must the system be in before the use case?):

- The user is in the application.
- The user hits the button to see all the available clothing in the system
- User completed use case 2 at least once

Main Flow (The main flow is a series of declarative steps):

1. User selects clothing item to generate outfit around
2. Item is matched to outfits that include the item
3. System sends the application 3 different outfit choices
4. User is shown 3 different outfit options
5. User clicks select outfit button
6. The application gets selected outfit from system
7. The system matches selected outfit to application
8. The application shows outfit on user

Subflows(Sub-flows break down the main flow and other sub-flows to improve document readability.):

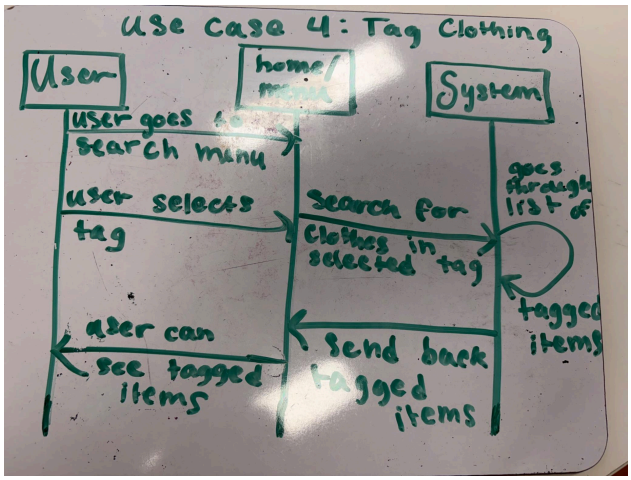
1. None

Alternative Flows (The alternative flows define exceptional behavior that can interrupt the normal flow. Often alternative flows indicate what is to be done under error conditions.

To determine alternative flows, ask yourself, "What could possibly go wrong?" for each of the actions in the main flow and the sub-flow):

1. No outfits can be made because a clothing item is missing
2. Less than three outfits are possible

UC4: Search Tagged Clothes



Preconditions:

- **User completed at least use case 1 once**
- **User has correctly tagged all clothes in their closet**

Main Flow

1. **The user goes to the search bar at the top of the page**
2. **The user goes to tag part of search bar**
3. **The user selects what type of clothes they want (i.e short sleeves, long sleeves, pants, etc)**
4. **The user selects the clothes that they want to wear**

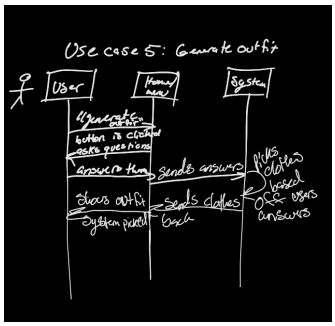
Subflows

1. None

Alternative Flows (exception/edge cases)

- **No clothes are found with that tag, so the program prints out "No items found"**
- **Clothes with selected tag already have been worn**

UC5: Generate Outfit



Preconditions:

- **User completed at least use case 1 once**
- **User has at least one full outfit in the closet (i.e top, bottom, dress)**

Main Flow

1. **User opens the program**
2. **User hits "generate outfit" button**
3. **User answers 3 basic questions [s1]**
4. **System takes answers and returns outfit**

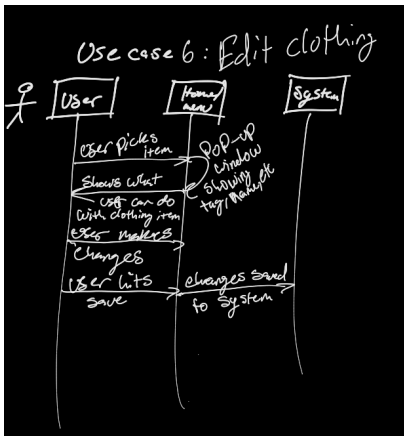
Subflows

1. **User answers these basic questions are 1. "Do you want to wear masculine or feminine style?" 2. "Is it hot or cold outside today?"**

Alternative Flows (exception/edge cases)

- **If all the users' clothes are dirty, the system wouldn't have any clean clothes to choose from. The system shows a message saying "All clothes are dirty, Please clean them or add new ones"**

UC6: Edit Clothing



Preconditions:

- **User completed use case 1 at least once**

Main Flow

- **User selects item item of clothing that they want to edit**
- **The clothing item pops up, showing the tag, name, and if they want to delete the clothing item.**
- **User can then change the name, tag, or picture of the item**
- **User clicks save and the changes made are now saved**

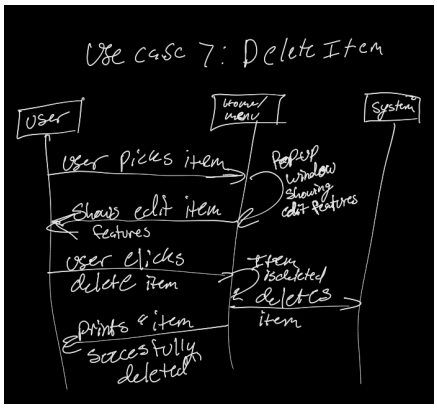
Subflows

- **None**

Alternative Flows (exception/edge cases)

- **None**

UC7: Delete Clothing



Preconditions:

- **User completed use case 1 and 6 at least once**

Main Flow

1. **User clicks on the item that they want to delete**
2. **The edit window pops up like in use case 6**
3. **At the bottom of the window screen has a delete button**
4. **Once user clicks on it, the item is deleted from the closet**

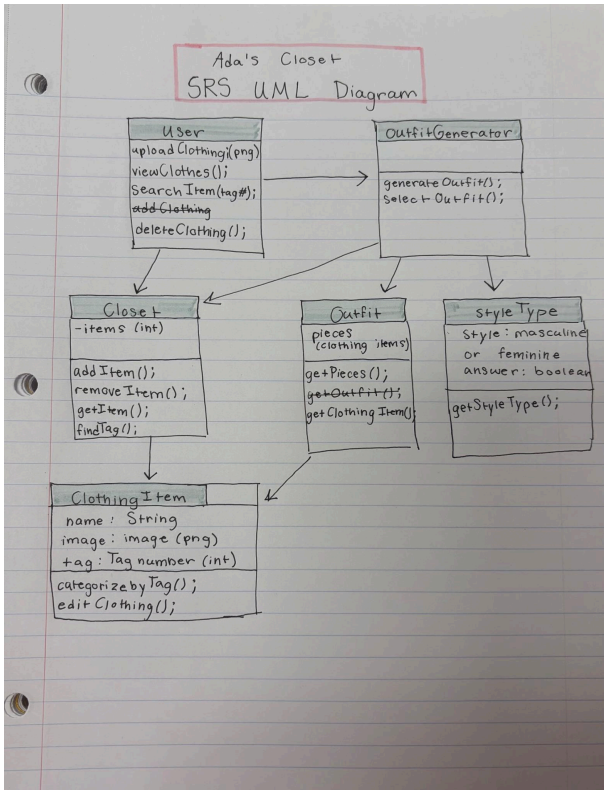
Subflows

- **None**

Alternative Flows (exception/edge cases)

- **None**

UML Diagram


Ada's Closet
SRS UML Diagram
User
uploadClothing(png)
viewClothes();
SearchItem(tag#);
~~addClothing~~
deleteClothing();
OutfitGenerator
generateOutfit();
selectOutfit();
Closet
-items (int)
addItem();
removeItem();
getItem();
findTag();
Outfit
pieces (clothing items)
getPieces();
~~getOutfit();~~
getClothingItem();
StyleType
Style: masculine or feminine
answer: boolean
getStyleType();
ClothingItem
name: String
image: image (png)
tag: Tag number (int)
categorizebyTag();
editClothing();

- Accessibility
    - Color contrast
    - Large text
    - Screen reader friendly
    - Accessibility fonts
    - Language options
- Security
- Software responds quickly (10s)

- Time limit: We may run out of time to work on the project and not get all the requirements in
- AI use: In its most complete form Ada's closet would implement AI to scan the clothing item and analyze its color, type, and occasion to waer it too.
- Resources: We dont have the correst resources to complete the project
- Application based: Not having the application working
- Experience: Not having the all the experience that we need to complete the project to our likings

- Visual studio code
- IOS
- Android

The program will be delivered either via website or application. Allowing the user to have the program choose or help choose an outfit for the day.